



“Tobby”, “swabwee”, and “strawberry”: Speech productions by toddlers in a linguistically-diverse community

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Background

- Young children vary considerably in their speech production patterns and intelligibility to adults
- Despite its relevance for clinical assessments, little is known about how input diversity, including multilingualism & accent variation, shapes speech development

Hypothesis: Diverse input causes additional unexpected deviations from adult-like forms, decreasing intelligibility

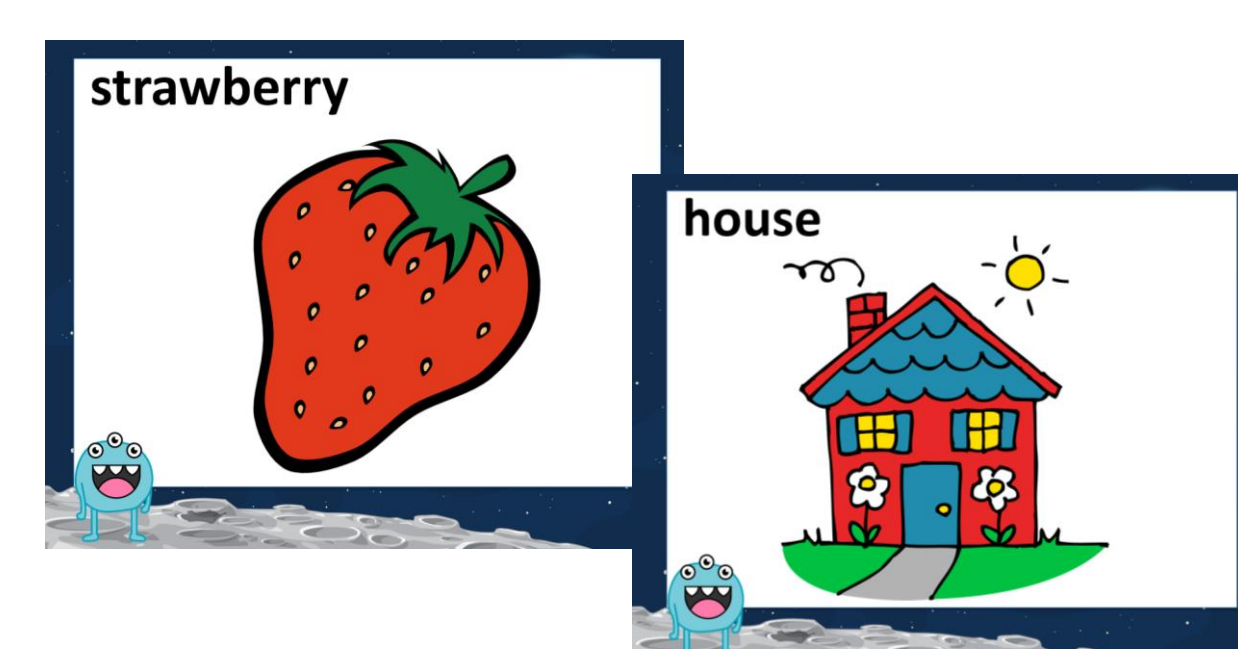
Method

Toddler production study

Participants: Age 2.25-3.25 years (27-39 months)

- Monolingual + Monoaccent** ($n = 28$)
- Monolingual + Multiaccent** ($n = 28$): Routinely hear both Canadian English and other accent/dialect
- Bilingual** ($n = 28$): Hear English 50-70% of the time

Task: Picture elicitation (must contribute ≥ 16 of 22 items)



Adult perception study

Participants: Undergraduate students (current $n = 43$)

Task: Open-set transcription, administered on Gorilla


Replay once
Guess the word →

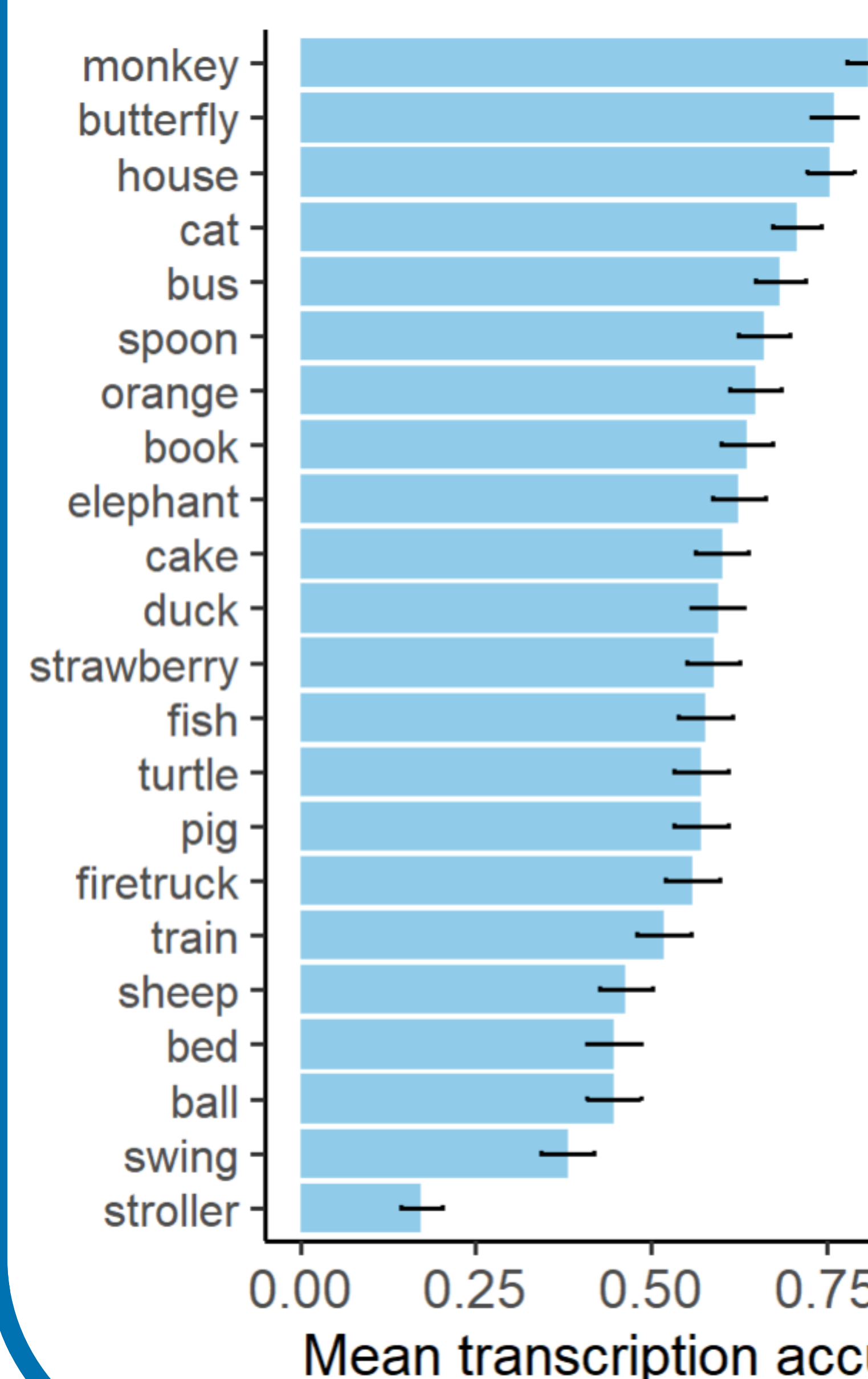
Make your best guess

Next

x 176 trials
88 target
= 22 items x 4 productions
88 filler

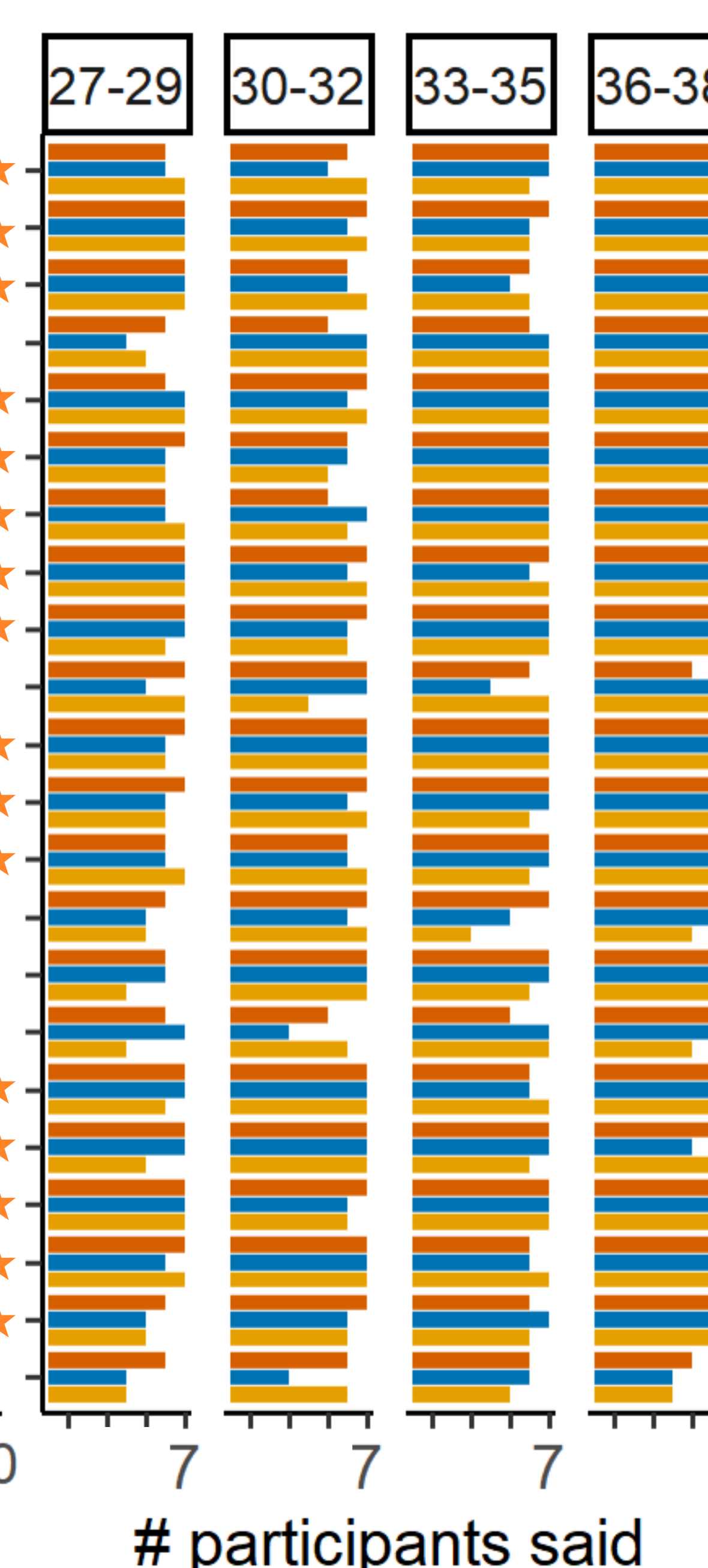
Results

Which words were most intelligible?

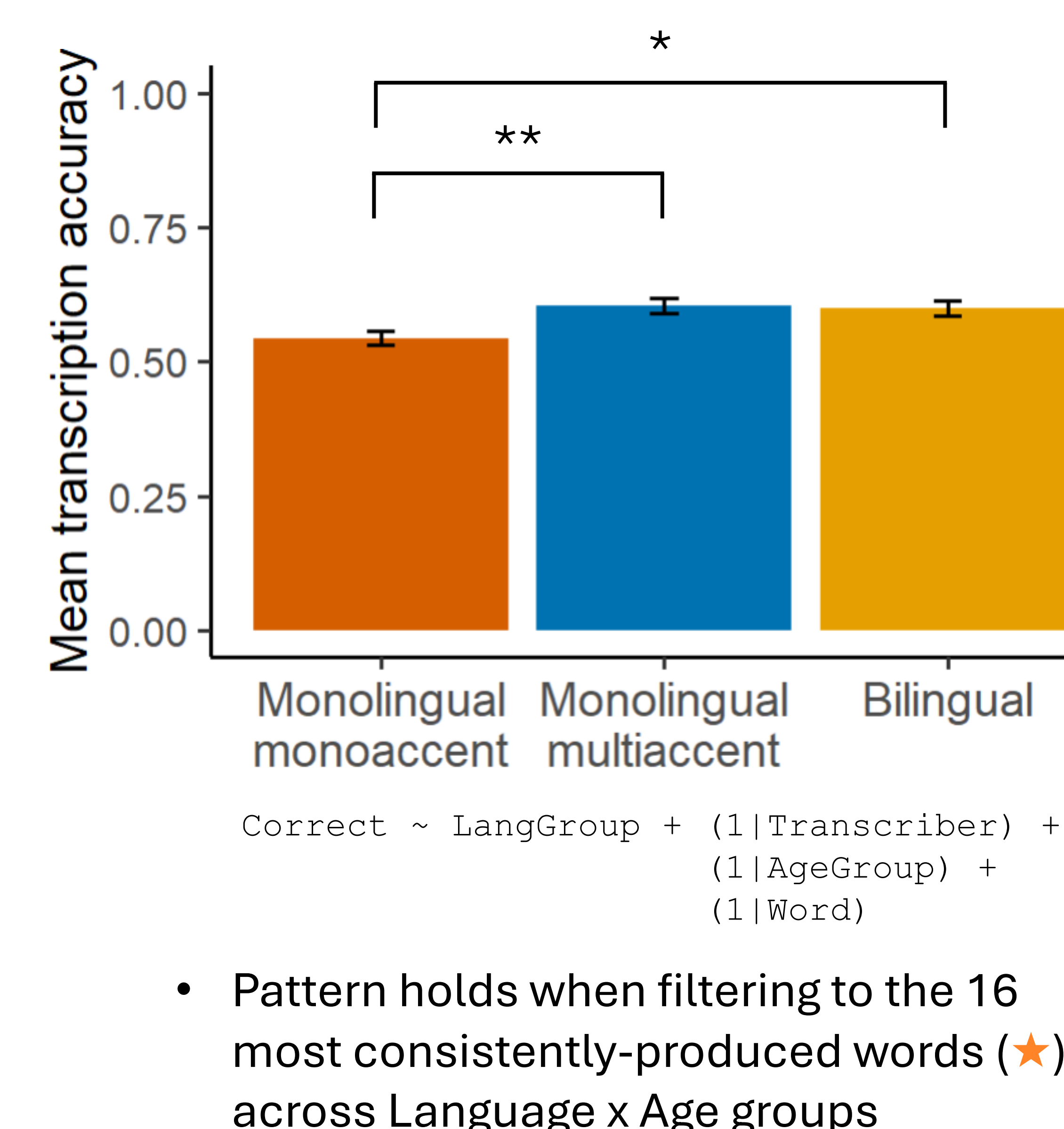


Who said them?

Age Group (months)



Overall, which groups were most intelligible?



Discussion

- Contrary to hypotheses, preliminary data suggests toddlers with high-diversity input are **more** intelligible
- Limitation: Inclusion criteria may select for children with advanced speech abilities, particularly in bilingual & multiaccent groups
- Ongoing work includes acoustic analyses of toddler productions across language groups, and consideration of parent-reported vocabulary

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